MERCK MANUAL
Professional Version

Asthma

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Asthma is a disease characterized by diffuse airway inflammation caused by a variety of triggering stimuli resulting in partially or completely reversible bronchoconstriction. Symptoms and signs include dyspnea, chest tightness, cough, and wheezing. The diagnosis is based on history, physical examination, and pulmonary function tests. Treatment involves controlling triggering factors and pharmacotherapy, most commonly with inhaled beta-2 agonists and inhaled glucocorticoids. Prognosis is good with treatment.

Asthma is a common chronic respiratory disease that is noncommunicable. It is a heterogenous disease and usually characterized by chronic airway inflammation and hyperresponsiveness (1). It is defined by a history of respiratory symptoms such as wheeze, shortness of breath, chest tightness, and cough that vary over time and in intensity, together with variable expiratory airflow limitation. (See also Wheezing and Asthma in Infants and Young Children.)

General reference

1. Global Initiative for Asthma. Global Strategy for Asthma Management and Prevention, 2025. Updated May 2025. Accessed May 20, 2025. Available from www.ginasthma.org

Epidemiology of Asthma

The global prevalence of asthma varies significantly among different regions and populations, influenced by factors such as socioeconomic status, environmental exposures, and genetic predisposition. One meta-analysis estimated that 260 million individuals worldwide were affected by asthma in 2021, with the age-adjusted prevalence rate of asthma being 3340 per 100,000 people worldwide (1). Asthma accounts for around 420,000 deaths each year globally (2).

The United States has higher asthma prevalence rates than most other countries worldwide. A meta-analysis of global asthma trends revealed that the United States had the highest age-adjusted prevalence rate at 10,400 per 100,000 people (3). Another study of NHANES (National Health and Nutrition Examination Survey) data in adults over 20 years old between 1999 and 2020 reported an overall asthma prevalence of 8.7% (4). Approximately 25 million people in the United States are estimated to be affected (5). Asthma occurs more frequently in non-Hispanic Black people and people of Puerto Rican ancestry. In adults, asthma is more commonly reported in women (6).

In the United States, about 10,000 deaths occur annually as a result of asthma, and the mortality rate is declining (7). However, the death rate is 2 to 3 times higher for Black patients than for White patients. The mortality rate is higher for adults than children and is especially high in adults over 65 years. Asthma is estimated to cost the United States $82 billion/year in medical care and lost productivity (8).

Asthma is one of the most common chronic diseases of childhood, affecting about 6.5% (4.6 million) of children in the United States (9). Asthma is among the leading causes of hospitalization in children and is a leading cause of school absenteeism (10). In contrast to adults, asthma in childhood has a male predominance (11).

Epidemiology references

Etiology of Asthma

The development of asthma is multifactorial and depends on the interactions among multiple susceptibility genes and environmental factors. Asthma may be classified into 2 broad endotypes based on immunopathologic mechanisms; these are **type 2 (T2)-high endotype** and **type 2 (T2)-low endotype**. T2-high endotypes are characterized by eosinophilic airway inflammation (sometimes also called eosinophilic asthma) and responsiveness to corticosteroids and biologics. T2-low endotypes are characterized by the absence of eosinophilic inflammation and may be further classified to include **neutrophilic type** or **paucigranulocytic type**, which are based on the presence or absence of neutrophils; both types are characterized by relative glucocorticoid resistance.

Numerous asthma susceptibility genes have been identified. Many are thought to involve the broad category of T-helper cells type 2 (Th2) and may play a role in inflammation. Examples include the *FCER1B* gene, which encodes the β-chain of the high-affinity IgE receptor; the genes encoding certain interleukins (IL) and their receptors, such as IL-4, IL-13, and the IL-4 receptor; genes responsible for innate immunity (HLA-DRB1, HLA-DQB1, CD14); and genes participating in cellular inflammation (eg, genes encoding granulocyte-monocyte colony-stimulating factor [GM-CSF] and tumor necrosis factor-alpha [TNF-α]). Also, the *ADAM33* gene, which encodes a protein that stimulates airway smooth muscle and fibroblast proliferation and remodeling, was the first asthma risk locus found with whole-genome family linkage studies.

Genome-wide association studies have identified multiple susceptibility loci as risk factors for developing asthma. The most replicated is at the chromosome 17q21 locus. This locus contains the *ORMDL3* gene, which is an allergen and cytokine (IL-4/IL-13)–inducible gene implicated in epithelial cell remodeling and sphingolipid metabolism to affect bronchial hyperreactivity. Another susceptibility locus is thymic stromal lymphopoietin, *TSLP*; single nucleotide polymorphisms in *TSLP* are responsible for initiating airway inflammation (1).

Epigenetic factors are also thought to contribute to the development of asthma. One epigenome-wide association study of blood DNA methylation levels in adults with non-atopic and atopic asthma identified numerous differentially methylated sites (ie, evidence of epigenetic modifications) for both conditions (2). Atopic asthma had evidence of greater relative methylation.

Environmental risk factors for asthma may include the following:

- Environmental exposures (allergens, infections, pollutants)
- Diet and obesity
- Perinatal factors
- Socioeconomic status

Genetic and environmental components may interact. Evidence clearly implicates household allergens (eg, dust mite, cockroach, pet) and other environmental allergens (eg, pollen) in disease development in older children and adults.

Several endotypes (subtypes of a disease defined by a distinct pathophysiological mechanism) of asthma exist. Infants may be born with a predisposition toward proallergic and proinflammatory type 2 (T2) immune responses (immune responses related to T-helper 2 cells). The hygiene hypothesis posits that early childhood exposure to bacterial and viral infections may shift the body toward T-helper cells type 1 (Th1) responses, which suppress Th2 cells and induce tolerance. Type 1 (T1) responses are characterized by a proliferation of Th1 cells. Trends toward having smaller families with fewer children, cleaner indoor environments, and early use of vaccinations and antibiotics may deprive children of these T2-suppressing, tolerance-inducing exposures and may partly explain the increase in asthma prevalence in higher income countries. Endotoxin exposure (from lipopolysaccharide components in the outer membrane of some Gram-negative bacteria) early in life can induce tolerance and may be protective. The body's microbiome also plays an important role in regulating immune responses. A diverse microbiome is thought to promote a balanced immune response, whereas a lack of microbial diversity (often prevalent in highly sanitized indoor environments) can lead to dysregulation of immune responses that contribute to asthma development (3). Air pollution is not definitively linked to asthma development, although it may trigger exacerbations.

Diets low in vitamins C and E and in omega–3 fatty acids have been linked to asthma; however, several studies supporting dietary influence are limited by sample size or did not account for differences in socioeconomic, environmental, and demographic factors. Dietary supplementation with these substances does not appear to prevent asthma. Asthma has also been linked to perinatal factors, such as young maternal age, poor maternal nutrition, prematurity, low birthweight, and lack of breastfeeding.

Obesity is considered an important modifiable risk factor for asthma and often precedes the diagnosis of asthma. Key mediators implicated by observational and cross-sectional studies include leptin, adipokines, and serum interleukin (IL)-6. However, underlying mechanisms are not known. Multiple studies have shown decreases in asthma severity and exacerbations after weight loss (4).

Lower socioeconomic status, active smoking, and a family history of asthma have also been found to be risk factors for the development of asthma (5).

Reactive airways dysfunction syndrome (RADS) and irritant-induced asthma

Reactive airways dysfunction syndrome (RADS) is the rapid onset (minutes to hours, but not > 24 hours) of an asthma-like syndrome that

- Develops in people with no history of asthma
- Occurs following a single, specific inhalation exposure to a significant amount of an irritating gas or particulate
- Persists for ≥ 3 months

Numerous substances have been implicated, including chlorine gas, nitrogen oxide, and volatile organic compounds (eg, from paints, solvents, adhesives). The exposure event is usually obvious to the patient, particularly when symptoms begin almost immediately.

Irritant-induced asthma refers to a similar, persistent asthma-like response following multiple or chronic inhalational exposure to high levels of similar irritants. Manifestations are sometimes more insidious, and thus the connection to the inhalational exposure is clear only in retrospect. See Work-Related Asthma for more information.

RADS and chronic irritant-induced asthma have many clinical similarities to asthma (eg, wheezing, dyspnea, cough, presence of airflow limitation, bronchial hyperresponsiveness) and respond significantly to bronchodilators and often glucocorticoids. Unlike in asthma, the reaction to the inhaled substance is not thought to be an IgE-mediated immune response; low-level exposures do not cause either RADS or irritant-induced asthma. However, repeated exposure to the initiating agent may trigger additional symptoms.

Etiology references

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Pathophysiology of Asthma

Asthma involves:

- Bronchoconstriction
- Airway edema and inflammation
- Airway hyperreactivity
- Airway remodeling

In patients with asthma, T-helper 2 (Th2) cells (a subset of $CD4^+$ T lymphocytes), and other cell types—notably, eosinophils and mast cells, but also other $CD4^+$ T cells, neutrophils, and natural killer T cells —form an extensive inflammatory infiltrate in the airway epithelium and smooth muscle, leading to airway remodeling (ie, desquamation, subepithelial fibrosis, angiogenesis, and smooth muscle hypertrophy). Hypertrophy of smooth muscle narrows the airways and increases reactivity to triggers such as allergens, infections, irritants, and parasympathetic stimulation. Exposure to triggers then causes release of proinflammatory neuropeptides, such as substance P, neurokinin A, and calcitonin gene-related peptide, and other mediators (eg, inflammatory cytokines) implicated in bronchoconstriction.

The pathophysiology of T2-low asthma (also called non-T2), is complex and influenced by various factors, including viral infections, smoking, obesity, and environmental pollutants. T2-low endotypes include neutrophilic asthma and paucigranulocytic asthma (1). In contrast with the T2-high (eosinophilic) endotype, neutrophilic asthma is often associated with increased levels of Th1 and Th17 cytokines, such as IL-6 and IL-17, which promote neutrophil recruitment and activation. Neutrophil extracellular traps and inflammasome (multi-protein complex that acts as a key component of the innate immune system, triggering inflammatory responses) activation promote local airway inflammation and are implicated in the pathogenesis of severe neutrophilic asthma. Paucigranulocytic asthma is characterized by minimal granulocytic inflammation (ie, eosinophilic and neutrophilic) and uncoupling of airway inflammation, hyperresponsiveness, and eventual remodeling (2).

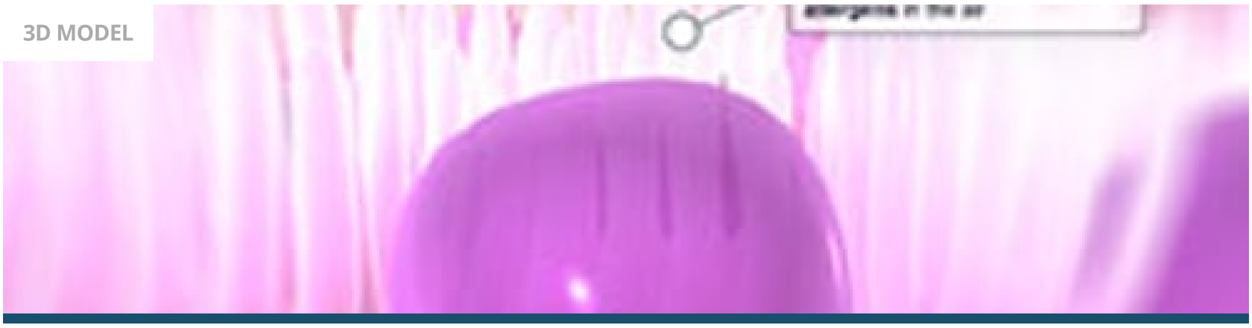
The primary cytokines involved in the pathogenesis of asthma are IL-4, IL-5, IL-13 (3). IL-4 promotes the differentiation of undifferentiated T cells into Th2 cells and induces B-lymphocyte immunoglobulin class switching to synthesis of IgE production. IL-4 also induces endothelial cell expression of adhesion molecules responsible for the recruitment of eosinophils, basophils, and T cells. IL-5 serves as the primary hematopoietic cytokine regulating eosinophil maturation and survival. IL-13 contributes to airway eosinophilia, mucus gland hyperplasia, airway fibrosis, and remodeling. Epithelial cells in patients with asthma often secrete increased amounts of alarmins, including thymic stromal lymphopoietin (TSLP) and IL-33. Alarmins activate Th2 cells and other innate immune cells. These pathways are clinically significant as they have been identified as therapeutic targets in the management of severe asthma.

Asthma occurring as a part of aspirin-exacerbated respiratory disease is characterized by a dysregulation of arachidonic acid metabolism.

Additional contributors to airway hyperreactivity include loss of inhibitors of bronchoconstriction (epithelium-derived relaxing factor, prostaglandin E2) and loss of other substances called endopeptidases that metabolize endogenous bronchoconstrictors. Mucus plugging and peripheral blood eosinophilia are additional classic findings in asthma and may be epiphenomena of airway inflammation. However, not all patients with asthma have eosinophilia.

Allergic Eosinophilic Asthma

3D MODEL



Asthma triggers

Common triggers of an asthma exacerbation include the following:

- Environmental and occupational allergens (numerous)
- Cold, dry air
- Infections
- Exercise
- Inhaled irritants
- Emotion
- Aspirin and other nonsteroidal anti-inflammatory drugs (NSAIDs)
- Gastroesophageal reflux disease (GERD)

Atopy (the genetic predisposition to form allergen-induced IgE responses) is the most significant identifiable risk factor for developing asthma (4). Allergic rhinitis often coexists with asthma; allergic triggers responsible for the inflammatory response in the upper airway (ie, rhinitis) may also be involved in causing similar responses in the lower airway (ie, asthma). Multiple studies have shown that rhinitis itself, even without allergen inhalation, can be an independent risk factor for asthma through shared mechanisms of airway inflammation and nasal-bronchial reflexes (5, 6).

Infectious triggers in young children include respiratory syncytial virus, rhinovirus, and parainfluenza virus infection. In older children and adults, upper respiratory infections (particularly with rhinovirus) and pneumonia are common infectious triggers. Exercise can be a trigger, especially in cold or dry environments, and cold air alone can also trigger bronchoconstriction. Inhaled irritants, such as air pollution, cigarette smoke, perfumes, and cleaning products can also trigger symptoms in patients with asthma. Inhaled irritants that trigger asthma exacerbations do so by inducing a T2 response, in contrast to what happens in reactive airways dysfunction syndrome and chronic irritant-induced asthma. Emotions such as anxiety, anger, and excitement sometimes trigger exacerbations.

Aspirin and other NSAIDs are triggers for exacerbations in up to 30% of patients with nasal polyps and in approximately 9% of all patients with asthma (7). Aspirin -exacerbated respiratory disease is typically accompanied by chronic rhinosinusitis with nasal polyposis. This condition is also known as **Samter's triad** (asthma, nasal polyposis, and sensitivity to aspirin and other cyclooxygenase-1 [COX-1]-inhibiting NSAIDs).

GERD is a common trigger among some patients with asthma, possibly via esophageal acid-induced reflex bronchoconstriction or by microaspiration of acid. However, treatment of asymptomatic GERD (eg, with proton pump inhibitors) does not seem to improve asthma control.

Response

In the presence of triggers, there is reversible airway narrowing and uneven lung ventilation. In lung regions distal to narrowed airways, relative perfusion exceeds relative ventilation; thus, alveolar oxygen tensions decrease and alveolar carbon dioxide tensions increase. Usually, regional hypoxia and hypercarbia trigger compensatory pulmonary vasoconstriction to match regional ventilation and perfusion; however, these compensatory mechanisms can fail during an asthma exacerbation due to the vasodilatory effects of prostaglandins that are upregulated during an exacerbation. Most patients can compensate by hyperventilating, but during severe exacerbations, diffuse bronchoconstriction may lead to severe air trapping. Bronchoconstriction coupled with air trapping places the respiratory muscles at a marked mechanical disadvantage so that the work of breathing increases. Under these conditions, hypoxemia worsens and $PaCO_2$ rises. Respiratory acidosis and metabolic acidosis may result and, if left untreated, cause respiratory and cardiac arrest.

Pathophysiology references

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Classification of Asthma

Asthma causes a number of clinical and testing abnormalities and manifestations typically wax and wane. Thus, because of variations in the clinical presentation, monitoring (and studying) asthma requires a consistent terminology and defined benchmarks.

Asthma can be classified in several ways, including based on severity (intermittent versus persistent), clinical phenotypes (eg, allergic, non-allergic, adult-onset) and mechanistic endotypes (T2-high, T2-low).

The term status asthmaticus describes severe, intense, prolonged bronchospasm that is resistant to treatment.

Severity

Severity is the intrinsic intensity of the disease process (ie, how bad it is—see table Classification of Asthma Severity). Severity can usually be assessed directly only before treatment is started, because patients who have responded well to treatment by definition have few symptoms. Asthma severity is categorized as:

- Intermittent
- Mild persistent
- Moderate persistent
- Severe persistent

It is important to remember that the severity category does not predict how serious an exacerbation a patient may have. For example, a patient who has mild asthma with long periods of no or mild symptoms and normal pulmonary function may have a severe, life-threatening exacerbation.

TABLE

Classification of Asthma Severity*

Components of Severity	Intermittent	Mild Persistent	Moderate Persistent	Severe Persistent
Symptoms and risk measures	All ages: ≤ 2 days/week	All ages: > 2 days/week, not daily	All ages: Daily	All ages: Throughout the day
Nighttime awakenings	Adults and children ≥ 5 years: ≤ 2 times/month	Adults and children ≥ 5 years: 3–4 times/month	Adults and children ≥ 5 years: > 1	Adults and children ≥ 5 years: Often 7 times/week

Components of Severity	Intermittent	Mild Persistent	Moderate Persistent	Severe Persistent
	Children 0–4 years: 0	Children 0–4 years: 1–2 times/month	time/week but not nightly Children 0–4 years: 3–4 times/month	Children 0–4 years: > 1 time/week
Reliever use†, ‡ for symptoms (not to prevent EIB)	≤ 2 days/week	Adults and children ≥ 5 years: > 2 days/week but not daily Children 0–4 years: > 2 days/week but not daily	Daily	Several times per day
Interference with normal activity	None	Minor limitation	Some limitation	Extreme limitation
FEV1	Adults and children ≥ 5 years: > 80% Children 0–4 years: Not applicable	Adults and children ≥ 5 years: > 80% Children 0–4 years: Not applicable	Adults and children ≥ 5 years: 60–80% Children 0–4 years: Not applicable	Adults and children ≥ 5 years: < 60% Children 0–4 years: Not applicable
FEV1/FVC	Adults and children ≥ 12 years: Normal† Children 5–11 years: > 85% Children 0–4 years: Not applicable	Adults and children ≥ 12 years: Normal† Children 5–11 years: > 80% Children 0–4 years: Not applicable	Adults and children ≥ 12 years: Reduced 5%† Children 5–11 years: 75–80% Children 0–4 years: Not applicable	Adults and children ≥ 12 years: Reduced > 5%† Children 5–11 years: < 75% Children 0–4 years: Not applicable
Risk of an asthma exacerbations requiring oral	0–1/year	Adults and children ≥ 5 years: ≥ 2/year	More frequent and intense events indicate greater severity	More frequent and intense events indicate greater severity

Components of Severity	Intermittent	Mild Persistent	Moderate Persistent	Severe Persistent
glucocorticoid bursts¶		Children 0–4 years: ≥ 2 in 6 months or wheezing ≥ 4 times/year lasting > 1 day AND risk factors for persistent asthma		

* Severity is categorized based on degree of impairment and risk of exacerbations requiring oral glucocorticoids. Impairment is assessed over the previous 2–4 weeks, and risk is assessed over the past year. After controller therapy is initiated, severity should be assessed by the level of treatment required to maintain adequate control.

† Evidence for airflow obstruction is based on an FEV1/FVC ratio less than expected normal values by age group. Normal FEV1/FVC ratios by age group: 8–19 years = 85%; 20–39 years = 80%; 40–59 years = 75%; 60–80 years = 70%.

‡ Relievers, also called rescue inhalers, include a short-acting beta-2-agonist, an inhaled glucocorticoid plus a short-acting beta-2-agonist, or an inhaled glucocorticoid plus a long-acting beta-2-agonist.

¶ At present, there are inadequate data to correlate frequencies of exacerbations with different levels of asthma severity. In general, more frequent and intense exacerbations (eg, requiring urgent, unscheduled care such as visits to the emergency department, hospitalization, or intensive care unit admission) indicate greater underlying disease severity. For treatment purposes, patients with ≥ 2 exacerbations in the preceding 12 months may be considered to have persistent asthma.

EIB = exercise-induced bronchoconstriction; FEV1 = forced expiratory volume in 1 second; FVC = forced vital capacity; ICS = inhaled corticosteroid (glucocorticoid); SABA = short-acting beta-2 agonist.

Adapted from National Heart, Lung, and Blood Institute: Expert Panel Report 3: Guidelines for the diagnosis and management of asthma—full report 2007. August 28, 2007. Available at http://www.nhlbi.nih.gov/guidelines/asthma/asthgdln.htm and Expert Panel Working Group of the National Heart, Lung, and Blood Institute (NHLBI) administered and coordinated National Asthma Education and Prevention Program Coordinating Committee (NAEPPCC), Cloutier MM, Baptist AP, et al: 2020 Focused Updates to the Asthma Management Guidelines: A Report from the National Asthma Education and Prevention Program Coordinating Committee Expert Panel Working Group [published correction appears in *J Allergy Clin Immunol* 2021 Apr;147(4):1528-1530. doi: 10.1016/j.jaci.2021.02.010]. *J Allergy Clin Immunol* 146(6):1217–1270, 2020. doi:10.1016/j.jaci.2020.10.003..

For additional information, see Global Initiative for Asthma. Global Strategy for Asthma Management and Prevention, 2025. Updated May 2025. Accessed May 20, 2025. Available from www.ginasthma.org

Control

Control is the degree to which symptoms, impairments, and risks are minimized by treatment. Control is the parameter assessed in patients receiving treatment. The goal is for all patients to have well-controlled asthma regardless of disease severity. Control is classified as:

- Well controlled
- Not well controlled
- Very poorly controlled

Severity and control are assessed in terms of patient impairment and risk (see tables Classification of Asthma Severity and Classification of Asthma Control).

TABLE

Classification of Asthma Control*,†

Component	Well Controlled	Not Well Controlled	Very Poorly Controlled
Symptoms	All ages except children 5–11 years: ≤ 2 days/week Children 5–11 years: ≤ 2 days/week but not > once/day	All ages except children 5–11 years: > 2 days/week Children 5–11 years: > 2 days/week or multiple times on ≤ 2 days/week	For all ages: Throughout the day
Nighttime awakenings	Adults and children ≥ 12 years: ≤ 2/month Children 5–11 years: ≤ 1 /month Children 0–4 years: ≤ 1/month	Adults and children ≥ 12 years: 1–3/week Children 5–11 years: ≥ 2/month Children 0–4 years: > 1/month	Adults and children ≥ 12 years: ≥ 4/week Children 5–11 years: ≥ 2/week Children 0–4 years: > 1/week
Interference with normal activity	None	Some limitation	Extreme limitation
Use of a reliever‡ for symptom control (not prevention of exercise-induced	≤ 2 days/week	> 2 days/week	Several times/day

Component	Well Controlled	Not Well Controlled	Very Poorly Controlled
bronchoconstriction)			
FEV1 or peak flow	> 80% predicted/personal best	60–80% predicted/personal best	< 60% predicted/personal best
FEV1/FVC (children 5–11 years)	> 80%	75–80%	< 75%
Exacerbations requiring oral systemic glucocorticoids¶	0–1/year	Adults and children ≥ 5 years: ≥ 2/year Children 0–4 years: 2–3/year	Adults and children ≥ 5 years: ≥ 2/year Children 0–4 years:> 3/year
Validated questionnaires:			
• ATAQ	0	1–2	3–4
• ACQ	≤ 0.75†	≥ 1.5	N/A
• ACT	≥ 20	16–19	≤ 15
Recommended action	Maintain current step Follow up every 1–6 months Consider step down if well controlled for ≥ 3 months	Step up 1 step Reevaluate in 2–6 weeks For adverse effects, consider treatment options	Consider short course of systemic glucocorticoids Step up 1 or 2 steps Re-evaluate in 2 weeks For adverse effects, consider treatment options

* All ages unless specified differently.

† Relievers, also called rescue inhalers, include a short-acting beta-2-agonist, an inhaled glucocorticoid plus a short-acting beta-2-agonist, or an inhaled glucocorticoid plus a long-acting beta-2-agonist.

‡ Level of control is based on the most severe impairment or risk category. Additional factors to consider are progressive loss of lung function on pulmonary function tests, significant adverse effects, and severity and interval between exacerbations (ie, one exacerbation requiring intubation or 2 hospitalizations within 1 month may be considered very poor control).

¶ At present, there are inadequate data to correlate frequencies of exacerbations with different levels of asthma control. In general, more frequent and intense exacerbations (eg, requiring urgent, unscheduled care such as visits to the emergency department, hospitalization, or intensive care unit admission) indicate poorer asthma control.

ACQ = asthma control questionnaire; ACT = asthma control test; ATAQ = asthma therapy assessment questionnaire; FEV1 = forced expiratory volume in 1 second; FVC = forced vital capacity.

Adapted from National Heart, Lung, and Blood Institute: Expert Panel Report 3: Guidelines for the diagnosis and management of asthma—full report 2007. August 28, 2007. Available at http://www.nhlbi.nih.gov/guidelines/asthma/asthgdln.htm and Expert Panel Working Group of the National Heart, Lung, and Blood Institute (NHLBI) administered and coordinated National Asthma Education and Prevention Program Coordinating Committee (NAEPPCC), Cloutier MM, Baptist AP, et al: 2020 Focused Updates to the Asthma Management Guidelines: A Report from the National Asthma Education and Prevention Program Coordinating Committee Expert Panel Working Group [published correction appears in *J Allergy Clin Immunol* 2021 Apr;147(4):1528-1530. doi: 10.1016/j.jaci.2021.02.010]. *J Allergy Clin Immunol* 146(6):1217–1270, 2020. doi:10.1016/j.jaci.2020.10.003.

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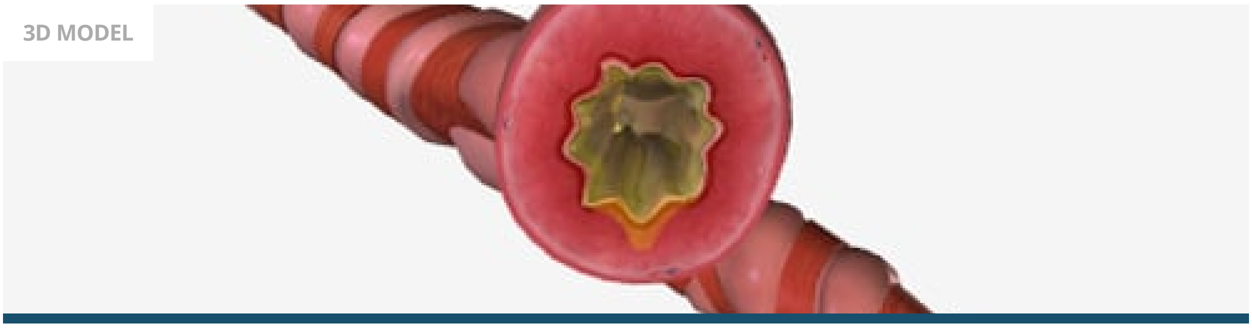
Asthma Attack

3D MODEL



Impairment

Impairment refers to the frequency and intensity of patients' symptoms and functional limitations (see table Classification of Asthma Severity). Impairment is assessed using similar criteria to severity, but differs from severity by its emphasis on symptoms and functional limitations rather than the intrinsic intensity of the disease process. Lung function or physiologic, objective impairment can be measured by spirometry, mainly forced expiratory volume in 1 second (FEV1) and the ratio of FEV1 to forced vital capacity (FVC), which correlate strongly with subjective components of asthma control that includes symptoms and clinical features such as the following:

- How often symptoms are experienced
- How often the patient awakens at night
- How often the patient uses a reliever (rescue inhaler) for symptom relief
- How often asthma interferes with normal activity

Risk

Risk refers to the likelihood of future exacerbations or decline in lung function and the risk of adverse medication effects. Risk is assessed by long-term trends in spirometry and clinical features such as:

- Frequency of need for oral glucocorticoids
- Need for hospitalization
- Need for intensive care unit (ICU) admission
- Need for intubation

Symptoms and Signs of Asthma

The clinical features of asthma can range in intensity from mild to severe. All symptoms and signs are nonspecific, are usually reversible with timely treatment, and are usually brought on by exposure to one or more triggers. Patients with mild asthma are typically asymptomatic between exacerbations. Patients with more severe disease and those with exacerbations experience dyspnea, chest tightness, audible wheezing, and coughing. Coughing may be the only symptom in some patients (cough-variant asthma).

Symptoms can follow a circadian rhythm and worsen during sleep, often around 4 AM. Many patients with more severe disease experience nighttime awakenings (nocturnal asthma).

Signs include wheezing, pulsus paradoxus (ie, a fall of systolic blood pressure [BP] > 10 mm Hg during inspiration), tachypnea, tachycardia, and visible efforts to breathe (use of neck and suprasternal [accessory] muscles, upright posture, pursed lips, speech limited by dyspnea). When severe, the expiratory phase of respiration is prolonged, with an inspiratory:expiratory ratio of at least 1:3. Wheezes can be present through both phases of respiration (severe) or just on expiration (mild). Patients with severe bronchoconstriction may have no audible wheezing because of markedly limited airflow, which is sometimes referred to as a silent chest and may indicate disease progression or impending respiratory failure.

Patients with a severe exacerbation and impending respiratory failure typically have some combination of altered consciousness, cyanosis, pulsus paradoxus > 15 mm Hg, oxygen saturation < 90%, PaO2 < 60 mm Hg, and PaCO2 > 45 mm Hg. Chest radiographs reveal hyperinflation often and pneumothorax or pneumomediastinum rarely. It is important to note that in patients with dark skin who have hypoxemia, pulse oximeters may overestimate oxygen saturation.

Symptoms and signs may disappear between exacerbations, although soft wheezes may be audible during forced expiration at rest, or after exercise, in some asymptomatic patients. Hyperinflation of the lungs may alter the chest wall in patients with long-standing uncontrolled asthma, causing a barrel-shaped thorax.

Diagnosis of Asthma

- History and physical examination
- Confirmatory pulmonary function testing
- Sometimes other tests

Diagnosis is suspected based on history and physical examination and is usually confirmed with pulmonary function tests (1). Diagnosis of causes and the exclusion of other disorders that cause wheezing are important.

Asthma and chronic obstructive pulmonary disease (COPD) are sometimes easily confused; they cause similar symptoms and produce similar results on pulmonary function tests but differ in important biologic ways that are not always clinically apparent. The T2-high endotype, because of allergic inflammation, is most commonly characterized by elevated exhaled nitric oxide, blood eosinophil counts, and serum IgE and is the most commonly recognized asthma subgroup. The T2-low endotype (also T1 cell-mediated immunity) is associated with elevated interferon-gamma, tumor necrosis factor, and neutrophilic inflammation; these findings have traditionally been associated with COPD but can occur in asthma subgroups not driven by T2 inflammation. These biologic mechanisms are not exclusive to either disease and can overlap between asthma and COPD.

Asthma-COPD overlap (ACO) is a unique entity that presents with persistent airflow obstruction and several features of both asthma and COPD. Key features include fixed airway obstruction not responsive to bronchodilators, significant exposure to tobacco smoking or pollutants, and traditional asthma features, including blood or sputum eosinophilia and reversible airflow obstruction. ACO represents an important subset of patients with asthma (up to 25%) and COPD (up to 33%) that might respond to medications not typically indicated for the disorder corresponding to the patient's main diagnosis (eg, prescribing roflumilast and azithromycin in a patient diagnosed with asthma or T2 biologic therapies in a patient diagnosed with COPD) (2).

Asthma that is difficult to control or refractory to commonly used controller therapies should be further evaluated for alternative causes of episodic wheezing, cough, and dyspnea such as allergic bronchopulmonary aspergillosis, bronchiectasis, asthma-COPD overlap, alpha-1 antitrypsin deficiency, cystic fibrosis, vocal fold dysfunction, and heart failure.

Pulmonary function tests

Patients suspected of having asthma should undergo pulmonary function testing to confirm and quantify the severity and reversibility of airway obstruction. Pulmonary function data quality is effort-dependent and requires patient instruction before the test. If it is safe to do so, bronchodilators should be discontinued before the test: 4 hours for short-acting beta-2 agonists (eg, albuterol); 24 hours for long-acting beta-2 agonists (eg, salmeterol, formoterol); 36 to 48 hours for ultra-long acting beta-2 agonists (eg, indacaterol, vilanterol); 12 hours for muscarinic antagonists that are short-acting (eg, ipratropium); and 48 hours for long-acting ones (eg, tiotropium).

Spirometry should be performed before and after inhalation of a short-acting bronchodilator. Signs of airflow limitation before bronchodilator inhalation include reduced FEV1 and a reduced FEV1/FVC ratio. The FVC may also be decreased because of air trapping, such that lung volume measurements may show an increase in the residual volume, the functional residual capacity, or both. A bronchodilator response should be determined by the following equation:

$$Bronchodilator\ response = \frac{(Post - bronchodilator\ value\ (L)\ -\ Pre - bronchodilator\ value\ (L))\ \times\ 100}{Predicted\ value\ (L)}$$

where predicted value is either FEV1 or FVC. A change of > 10% is considered a positive bronchodilator response (3). (Absence of a positive bronchodilator response should not preclude a therapeutic trial of long-acting bronchodilators.

Flow-volume loops should also be reviewed to diagnose vocal fold dysfunction, a common cause of upper airway obstruction that mimics asthma. However, vocal fold dysfunction is intermittent, and normal flow-volume loops do not exclude this condition.

Provocative testing, in which inhaled methacholine (or alternatives, such as inhaled histamine, adenosine, or bradykinin, or exercise testing) is used to provoke bronchoconstriction, is indicated for patients suspected of having asthma who have normal findings on spirometry and flow-volume testing and for patients suspected of having cough-variant asthma, provided there are no contraindications. Contraindications include FEV1 < 1 L or < 50% predicted, recent myocardial infarction or stroke, and severe hypertension (systolic BP > 200 mm Hg; diastolic BP > 100 mm Hg). A decline in FEV1 of ≥ 20% (> 12% in children) from baseline on a provocative testing protocol is relatively specific for the diagnosis of asthma (4). However, FEV1 may decline in response to medications used in provocative testing in other disorders, such as COPD. If FEV1 decreases by < 20% by the end of the testing protocol, asthma is less likely to be present.

Other tests

Other tests may be helpful in some circumstances:

- Diffusing capacity for carbon monoxide (DLCO)
- Chest radiography
- Allergy testing
- Fractional exhaled nitric oxide (FeNO)

DLCO testing can help distinguish asthma from COPD. Values are normal or slightly elevated in asthma and usually reduced in COPD, particularly in patients with emphysema.

A chest radiograph may help exclude some causes of asthma or alternative diagnoses, such as heart failure or pneumonia. The chest radiograph in asthma is usually normal but may show hyperinflation or segmental atelectasis, a sign of mucous plugging. Infiltrates, especially those that come and go and that are associated with findings of central bronchiectasis, suggest allergic bronchopulmonary aspergillosis.

Allergy testing in the skin or blood may be indicated for children whose history suggests allergic triggers (particularly for allergic rhinitis) because these children may benefit from immunotherapy. It should be considered for adults whose history indicates relief of symptoms with allergen avoidance and for those in whom a trial of therapeutic anti-IgE antibody therapy is being considered. Skin testing and measurement of allergen-specific IgE via radioallergosorbent testing (RAST) can identify specific allergic triggers.

Blood tests may be indicated. Elevated blood eosinophils (>150 to 300 cells/mcL [> 0.15×10^9 /L to 0.3×10^9/L]) and elevated nonspecific IgE levels are suggestive but are neither sensitive nor specific for a diagnosis of allergic asthma. Furthermore, some studies have indicated that eosinophil levels may vary diurnally and with other factors (5). Generally, blood eosinophil levels are higher in the morning, and there may be falsely low eosinophil counts when samples are collected in the afternoon. The diurnal fluctuations in eosinophil levels highlight the importance of carefully timing sample collection and conducting multiple measurements to ensure accurate asthma diagnosis and management.

In patients ≥ 5 years who are glucocorticoid-naive, FeNO may be used in the evaluation when the diagnosis of asthma is unclear, especially in children, and it can be used as a biomarker to monitor disease severity and therapeutic efficacy (6). FeNO levels > 50 ppb are consistent with allergic airways inflammation, supporting an asthma diagnosis. A level < 25 ppb is more consistent with an alternative diagnosis. Levels between 25 and 50 ppb are indeterminate. Treatment with glucocorticoids reduces eosinophilic airway inflammation; therefore, FeNO thresholds are lower in patients taking glucocorticoids (4). For patients treated with medium-dose inhaled glucocorticoids, a FeNO level ≥ 25 ppb is considered consistent with allergic airway inflammation; for patients treated with high-dose inhaled glucocorticoids, a FeNO level ≥ 20 ppb is considered consistent.

Sputum evaluation for eosinophils is not commonly done; finding large numbers of eosinophils is suggestive of asthma but not sensitive or specific.

Peak expiratory flow (PEF) measurements with handheld flow meters are recommended for home monitoring of disease severity and for guiding therapy.

CLINICAL CALCULATORS

Peak Expiratory Flow Prediction

Evaluation of exacerbations

Patients with asthma with an acute exacerbation are evaluated based on clinical criteria but should sometimes also have certain tests:

- Pulse oximetry
- Sometimes peak expiratory flow (PEF) measurement
- FeNO

The decision to treat an exacerbation is based primarily on an assessment of signs and symptoms. PEF measures can help establish the severity of an exacerbation but are most commonly used to monitor response to treatment in outpatients. PEF values are interpreted in light of the patient's personal best, which may vary widely among patients who are equally well controlled. A 15 to 20% reduction from this baseline indicates a significant exacerbation. When baseline values are not known, the percent predicted PEF based on age, height, and sex may be used, but this is less accurate than a comparison to the patient's personal best.

Although spirometry (eg, FEV1) reflects airflow more accurately than PEF, it is impractical in most urgent outpatient and emergency department settings (ie, in patients with a severe exacerbation needing emergent treatment). It may be used for office-based monitoring of treatment or when objective measures are required (eg, when an exacerbation appears to be more severe than perceived by the patient or is not recognized).

Chest radiograph is not necessary for most exacerbations but should be performed in patients with symptoms or signs suggestive of pneumonia, pneumothorax, or pneumomediastinum.

Arterial or venous blood gas measurements should be initiated in patients with marked respiratory distress or symptoms and signs of impending respiratory failure.

FeNO measurement has been proposed as an adjunct in evaluation of asthma control in patients with confusing or unclear histories or clinical scenarios. Higher baseline FeNO levels are linked to improved asthma control and reduced exacerbation rates. FeNO remains detectable during treatment with immunomodulatory medications, however, its dynamics while on such treatments require further investigation (7).

Diagnosis references

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Treatment of Asthma

- Control of triggers
- Pharmacotherapy
- Monitoring
- Patient education
- Treatment of acute exacerbations

Treatment objectives are to minimize impairment and risk, including preventing exacerbations and minimizing chronic symptoms, including nocturnal awakenings; to minimize the need for emergency department visits or hospitalizations; to maintain baseline (normal) pulmonary function and activity levels; and to avoid adverse treatment effects.

Control of triggering factors

Exposure to triggering factors in some patients may be controlled with use of synthetic fiber pillows and impermeable mattress covers and frequent washing of bed sheets, pillowcases, and blankets in hot water. Ideally, upholstered furniture, soft toys, carpets, curtains, and pets should be removed, at least from the bedroom, to reduce dust mites and animal dander. Dehumidifiers should be used in basements and in other poorly aerated, damp rooms to reduce mold. Steam cleaning of carpeting and upholstery reduces dust mite allergens. House cleaning and extermination to eliminate cockroach exposure are especially important. Although control of triggering factors is more difficult in urban environments, their significance remains unchanged.

High-efficiency particulate air (HEPA) vacuums and filters may relieve symptoms, but no beneficial effects on pulmonary function and on the need for medications have been observed.

Sulfite-sensitive patients should avoid sulfite-containing food (eg, certain wines and salad dressings).

Nonallergenic triggers, such as cigarette smoke, strong odors, irritant fumes, cold temperatures, and high humidity should also be avoided or controlled when possible. Limiting exposure to people with viral upper respiratory infections is also important. However, exercise-induced bronchoconstriction is not treated with exercise avoidance because exercise is important for health reasons. Instead, a short-acting bronchodilator is given prophylactically before exercise and as needed during or after exercise (reliever); controller therapy (step 2 and above in Table Steps of Asthma Management) should be started if exercise-induced symptoms are not responsive to reliever therapy or occur daily or more frequently.

Patients with aspirin-exacerbated respiratory disease should avoid cyclooxygenase-1 (COX-1) inhibitors, such as aspirin. When these patients need a pain reliever, they can use acetaminophen, choline magnesium salicylate, or a COX-2-specific inhibitor, such as celecoxib. Aspirin desensitization allows patients to tolerate aspirin and, by extension, COX-1 inhibition when maintained on a daily dose of aspirin post-desensitization.

Asthma is a relative contraindication to the use of nonselective beta-blockers (eg, propranolol, timolol, carvedilol, nadolol, sotalol), including topical formulations, but cardioselective medications (eg, metoprolol, atenolol) probably have no adverse effects.

Pharmacotherapy

Major medications classes commonly used in the treatment of asthma and asthma exacerbations include:

- Bronchodilators (beta or beta-2 adrenergic receptor agonists, anticholinergics)
- Glucocorticoids
- Leukotriene receptor antagonists
- Mast cell stabilizers
- Methylxanthines
- Biologic medications

Medications in these classes (see table Pharmacologic Treatment of Asthma) are inhaled, taken orally, or injected subcutaneously or intravenously; inhaled medications come in aerosolized and powdered forms. Use of aerosolized forms with a spacer or holding chamber facilitates deposition of the medication in the airways rather than the pharynx; patients are advised to wash and dry their spacers after each use to prevent bacterial contamination. In addition, use of aerosolized forms requires coordination between actuation of the inhaler (drug delivery) and inhalation; powdered forms reduce the need for coordination, because medication is delivered only when the patient inhales. For details, see Pharmacologic Treatment of Asthma.

Bronchial thermoplasty

Bronchial thermoplasty is a bronchoscopic technique in which heat is applied through a device that transfers localized controlled radiofrequency waves to the airways. The heat decreases the amount of airway smooth muscle remodeling (and thus the smooth muscle mass) that occurs with asthma. In both experimental and observational studies in patients with severe persistent asthma not controlled with multiple therapies, there have been modest decreases in exacerbation frequency and improvement in asthma symptom control (1, 2). However, some patients have experienced an immediate worsening of symptoms, sometimes requiring hospitalization immediately after the procedure. Expert recommendations are to avoid bronchial thermoplasty unless a patient places a low value on the potential for adverse outcomes and a high value on short-term potential benefits. If possible, bronchial thermoplasty should be done in centers with experience in the procedure (3).

Criteria for consideration of bronchial thermoplasty include severe asthma not controlled with inhaled glucocorticoids and long-acting beta agonists, intermittent or continuous use of oral glucocorticoids, FEV1 ≥ 50% of predicted, and no history of life-threatening exacerbations. Patients should understand the risk of post-procedure asthma exacerbation and need for hospitalization before proceeding with the procedure. There are limited data in patients with > 3 exacerbations per year or an FEV1 < 50% of predicted because these patients were excluded from the clinical trials (4).

Monitoring response to treatment

Guidelines recommend office use of spirometry (FEV1, FEV1/FVC, FVC) to measure airflow limitation and assess impairment and risk. Spirometry should be repeated at least every 1 to 2 years in patients with asthma to monitor disease progression. A step-up in therapy might be required if lung function declines or becomes impaired with evidence of increased airflow obstruction (see table Classification of Asthma Control). Outside the office, home peak expiratory flow (PEF) monitoring, in conjunction with patient symptom diaries and the use of an asthma action plan, is especially useful for charting disease progression and response to treatment in patients with moderate to severe persistent asthma. When asthma is quiescent, one PEF measurement in the morning suffices. Should PEF measurements fall to < 80% of the patient's personal best, then twice per day monitoring to assess circadian variation is useful. Circadian variation of > 20% indicates airway instability and the need to re-evaluate the therapeutic regimen (5).

Patient education

The importance of patient education cannot be overemphasized. Patients do better when they know more about asthma—what triggers an exacerbation, what medication to use when, proper inhaler technique, how to use a spacer with a metered-dose inhaler (MDI), and the importance of early use of glucocorticoids in exacerbations. Every patient should have a written action plan for day-to-day management, especially for management of acute exacerbations, that is based on the patient's best personal peak flow rather than on a predicted normal value. Such a plan leads to much better asthma control, largely attributable to improved adherence to therapies.

Treatment of acute asthma exacerbation

The goal of asthma exacerbation treatment is to relieve symptoms and return patients to their best lung function. Treatment includes:

- Inhaled bronchodilators (beta-2 agonists and anticholinergics)
- Usually systemic glucocorticoids

Details of the treatment of acute asthma exacerbations, including of severe attacks requiring hospitalization, are discussed elsewhere.

Treatment of chronic asthma

Asthma guidelines recommend treatment based on the severity classification (3, 6). Continuing therapy is based on assessment of control (see table Classification of Asthma Control). Therapy is increased in a stepwise fashion (see table Steps of Asthma Management) until the best control of impairment and risk is achieved (step-up). Before therapy is stepped up, adherence, exposure to environmental factors (eg, trigger exposure), and presence of comorbid conditions (eg, obesity, allergic rhinitis, gastroesophageal reflux disease, chronic obstructive pulmonary disease, obstructive sleep apnea, vocal fold dysfunction, inhaled cocaine use) are reviewed. These factors should be addressed before escalating (stepping up) medication therapy. Once asthma has been well controlled for at least 3 months, pharmacotherapy is reduced if possible to the minimum that maintains good control (step-down). For specific medications, see table Pharmacologic Treatment of Chronic Asthma.

TABLE

Steps of Asthma Management*

Step	Preferred Treatment†	Alternate Treatment
1 (starting point for intermittent asthma)	Low-dose inhaled glucocorticoid plus formoterol as needed† Short-acting beta-2 agonist as needed, along with low-dose inhaled glucocorticoid†	—
2 (starting point for mild persistent asthma)	Daily low-dose inhaled glucocorticoid and as-needed short-acting beta-2 agonist *or*	Mast cell stabilizer, leukotriene receptor antagonist, theophylline, or zileuton (for ages 12 and older), either with as-needed (rescue, or

Step	Preferred Treatment†	Alternate Treatment
	For ages 12 years and older, as-needed use of concomitant inhaled glucocorticoid and short-acting beta-2 agonist	reliever) short-acting beta-2 agonist therapy
3 (starting point for moderate persistent asthma)	Daily and as-needed combination therapy with low-dose inhaled glucocorticoid plus formoterol‡	Daily medium-dose inhaled glucocorticoid and as-needed (reliever) short-acting beta-2 agonist *or* Daily low-dose inhaled glucocorticoid plus one of the following: • Long-acting beta-2 agonist • Leukotriene receptor antagonist • Theophylline Each with as-needed (reliever) short-acting beta-2 agonist therapy *or* For ages 12 years and older, daily low-dose inhaled glucocorticoid plus a long-acting muscarinic antagonist or zileuton, either with as-needed (reliever) short-acting beta-2 agonist therapy
4	Daily and as-needed combination therapy with medium-dose inhaled glucocorticoid plus formoterol‡	Daily medium-dose inhaled glucocorticoid plus long-acting beta-2 agonist with as-needed (reliever) short-acting beta-2 agonist therapy *or*

Step	Preferred Treatment†	Alternate Treatment
		For ages 12 years and older, daily medium-dose inhaled glucocorticoid plus long-acting muscarinic antagonist with as-needed (reliever) short-acting beta-2 agonist therapy *or* Daily medium-dose inhaled glucocorticoid plus one of the following: • Leukotriene receptor antagonist • Theophylline • Zileuton (ages 12 and older) Each with as-needed (reliever) short-acting beta-2 agonist therapy
5 (starting point for severe persistent asthma)	For ages 5–11 years, daily high-dose inhaled glucocorticoid plus long-acting beta-2 agonist combination therapy with as-needed short-acting beta-2 agonist therapy *or* For ages 12 and older, daily medium-high dose inhaled glucocorticoid plus long-acting beta-2 agonist combination with long-acting muscarinic antagonist and as-needed short-acting beta-2 agonist therapy *and*	For ages 5–11 years, daily high-dose inhaled glucocorticoid plus leukotriene receptor antagonist or daily high-dose inhaled glucocorticoid plus theophylline, either with as-needed (reliever) short-acting beta-2 agonist therapy *or* For ages 12 and older, daily medium- to high-dose inhaled glucocorticoid plus long-acting beta-2 agonist or daily high-dose inhaled glucocorticoid plus

Step	Preferred Treatment†	Alternate Treatment
	Consider adding asthma biologics (includes anti-IgE, anti-IL5, anti-IL5R, anti-IL4/IL13)	leukotriene receptor antagonist, either with as-needed (rreliever) short-acting beta-2 agonist therapy *and* Consider adding asthma biologics (includes anti-IgE, anti-IL5, anti-IL5R, anti-IL-4/IL13)
6	Daily high-dose inhaled glucocorticoid plus long-acting beta-2 agonist plus oral glucocorticoid with as-needed short-acting beta-2 agonist therapy *and* Consider adding asthma biologics (includes anti-IgE, anti-IL5, anti-IL5R, anti-IL4/IL13)	For ages 5–11 years, daily high-dose inhaled glucocorticoid plus leukotriene receptor antagonist and oral systemic glucocorticoid or daily high-dose inhaled glucocorticoid plus theophylline plus oral glucocorticoid, either with as-needed (reliever) short-acting beta-2 agonist therapy *and* Consider adding asthma biologics (includes anti-IgE, anti-IL5, anti-IL5R, anti-IL4/IL13)

* Before stepping up, adherence, environmental factors (eg, trigger exposure), and comorbid conditions should be reviewed and managed if needed.

† A short-acting beta-2 agonist is indicated to provide quick relief at all steps and to prevent exercise-induced bronchoconstriction.

‡ The updated NAEPP Asthma Management Guidelines' preferred rescue option (also called reliever therapy) for steps 3 and 4 include the use of 1 to 2 puffs of as-needed formoterol in combination with an inhaled glucocorticoid, not to exceed a daily maximum dose of 8 puffs of formoterol (36 mcg) over a 24-hour period for ages 5–11 years, and 12 puffs of formoterol (54 mcg) over a 24-hour period for patients ≥ 12 years.

Adapted from the Expert Panel Working Group of the National Heart, Lung, and Blood Institute (NHLBI) administered and coordinated National Asthma Education and Prevention Program Coordinating Committee (NAEPPCC), Cloutier MM, Baptist AP, et al: 2020 Focused Updates to the Asthma Management Guidelines: A Report from the National Asthma Education and Prevention Program Coordinating Committee Expert Panel Working Group. *J Allergy Clin Immunol* 146(6):1217–1270, 2020. doi: 10.1016/j.jaci.2020.10.003

For additional information, see Global Initiative for Asthma. Global Strategy for Asthma Management and Prevention, 2025. Updated May 2025. Accessed May 20, 2025. Available from www.ginasthma.org. Note GINA does not recognize a step 6 for the management of severe asthma. Refer to guidelines for comparisons on the management of severe asthma for step 5 and above.

Exercise-induced bronchoconstriction

Exercise-induced bronchoconstriction can generally be prevented by prophylactic inhalation of a short-acting beta-2 agonist or mast cell stabilizer before starting the exercise. If beta-2 agonists are not effective or if exercise-induced bronchoconstriction causes symptoms daily or more frequently, the patient requires controller therapy.

Aspirin-exacerbated respiratory disease

The primary treatment for aspirin-exacerbated respiratory disease is avoidance of aspirin and other NSAIDs. Celecoxib does not appear to be a trigger. Leukotriene receptor antagonists can blunt the response to NSAIDs. Alternatively, desensitization can be done in either the inpatient or outpatient clinic setting depending on the severity of aspirin sensitivity and asthma severity; desensitization has been successful in the majority of patients who are able to continue desensitization treatment for more than one year.

Investigational therapies

Multiple therapies are being developed to target specific components of the inflammatory cascade. Therapies directed at interleukin 6 (IL-6), tumor necrosis factor-alpha, other chemokines, and cytokines or their receptors are all under investigation or consideration as therapeutic targets.

Treatment references

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Special Populations

Infants, children, and adolescents

Asthma is difficult to diagnose in infants; thus, under-recognition and undertreatment are common (see also Wheezing and Asthma in Infants and Young Children). Empiric trials of inhaled bronchodilators and anti-inflammatory medications may be helpful. Medications may be given by nebulizer or metered-dose inhaler (MDI) with a holding chamber with or without a face mask. Infants and children < 5 years who require treatment > 2 times/week should be given daily anti-inflammatory therapy with inhaled glucocorticoids (preferred), leukotriene receptor antagonists, or cromolyn. The nebulized route of administration is preferred in children under the age of 2 years.

Children > 5 years and adolescents with asthma should not receive long-acting muscarinic antagonists. Also, zileuton should only be used in children ≥ 12 years. Children > 5 years of age and adolescents with asthma should be encouraged to maintain physical activities, exercise, and sports participation. Predicted norms for pulmonary function tests in adolescents are closer to childhood (not adult) standards. Adolescents and mature younger children should participate in developing their own asthma management plans and establishing their own goals for therapy to improve adherence. The action plan should be understood by teachers and school nurses to ensure reliable and prompt access to rescue (reliever) medications. Cromolyn and nedocromil are often tried in this group but are not as beneficial as inhaled glucocorticoids. Long-acting medications may prevent the problems (eg, inconvenience, embarrassment) of having to take medications at school. However, older children and adolescents may be educated and instructed to carry albuterol for self-directed use prior to planned exercise to prevent exercise-induced bronchoconstriction.

Pregnant patients

Approximately 60% of pregnant patients with asthma notice worsening (at times to a severe degree), and 40% notice no change (1). Gastroesophageal reflux disease (GERD) may be an important contributor to symptomatic disease in pregnancy. Asthma control during pregnancy is crucial because poorly controlled maternal disease can result in increased prenatal mortality, premature delivery, and low birth weight.

Asthma medications have not been shown to have adverse fetal effects, but safety data are lacking. (In general, uncontrolled asthma is more of a risk to mother and fetus than adverse effects due to asthma medications. During pregnancy, normal blood PCO2 level is about 32 mm Hg (2). Therefore, carbon dioxide retention is probably occurring if PCO2 approaches 40 mm Hg.

Pearls & Pitfalls

- Suspect carbon dioxide retention and respiratory failure in pregnant women with uncontrolled asthma and PCO2 levels near 40 mm Hg.

Older patients

Older patients have a high prevalence of other obstructive lung disease (eg, COPD), so it is important to determine the magnitude of the reversible component of airflow obstruction (eg, by administering a 2- to 3-week trial of inhaled glucocorticoids or pulmonary function testing with bronchodilator challenge). Older patients may be more sensitive to adverse effects of beta-2 agonists and inhaled glucocorticoids. Patients requiring inhaled glucocorticoids, particularly those with risk factors for osteoporosis, may benefit from measures to preserve bone density (eg, calcium and vitamin D supplements, bisphosphonates).

Special populations references

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Prognosis for Asthma

Asthma resolves in many children. However, about 25% of children with clinically diagnosed asthma are at risk of accelerated declines in lung function (1). In a similar proportion of children, wheezing persists into adulthood or relapse occurs in later years (2). Female sex, smoking, earlier age of onset, and sensitization to household dust mites are risk factors for persistence and relapse.

Although a significant number of deaths each year are attributable to asthma, most of these deaths are preventable with treatment. Thus, the prognosis is good with adequate access and adherence to treatment. Risk factors for death include increasing requirements for oral glucocorticoids before hospitalization, previous hospitalization for acute exacerbations, and lower peak expiratory flow values at presentation. Several studies show that the use of inhaled glucocorticoids decreases hospital admission and mortality rates (3).

Over time, the airways in some patients with asthma undergo permanent structural changes (remodeling) and progress to baseline airflow obstruction that is not completely reversible. Early and aggressive use of anti-inflammatory medications may help prevent this remodeling.

Prognosis references

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Key Points

- **Asthma triggers range from environmental allergens and respiratory irritants to infections, aspirin, exercise, emotion, and gastroesophageal reflux disease.**
- **Consider asthma in patients who have unexplained persistent coughing, particularly at night.**
- **If asthma is suspected, arrange pulmonary function testing, with methacholine provocation if necessary.**
- **Educate patients on how to avoid triggers.**
- **Control chronic asthma with medications that modulate the allergic and immune response—usually inhaled glucocorticoids—with other medications (eg, long-acting bronchodilators, mast cell stabilizers, leukotriene receptor antagonists) added based on asthma severity.**

- **Treat asthma aggressively during pregnancy.**

More Information

The following English-language resources may be useful. Please note that The Manual is not responsible for the content of these resources.

Global Initiative for Asthma. Global Strategy for Asthma Management and Prevention, 2025. Updated May 2025. Accessed May 20, 2025. Available from www.ginasthma.org

The National Heart, Lung, and Blood Institute: Expert Panel Report 3, Guidelines for the diagnosis and management of asthma—full report 2007.

Expert Panel Working Group of the National Heart, Lung, and Blood Institute (NHLBI) administered and coordinated National Asthma Education and Prevention Program Coordinating Committee (NAEPPCC), Cloutier MM, Baptist AP, et al: 2020 Focused Updates to the Asthma Management Guidelines: A Report from the National Asthma Education and Prevention Program Coordinating Committee Expert Panel Working Group. *J Allergy Clin Immunol* 146(6):1217–1270, 2020. doi: 10.1016/j.jaci.2020.10.003

Drug Information for the Topic

MERCK MANUAL
Professional Version

Pharmacologic Treatment of Asthma

*Full Review: Dec 2025 By **Victor E. Ortega**, MD, PhD, Mayo Clinic | **Sergio E. Chiarella**, MD, Mayo Clinic | Peer reviewed by **M. Patricia Rivera**, MD, University of Rochester Medical Center*
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Asthma is a common chronic respiratory disease that is noncommunicable. The treatment of asthma is based on a stepwise, individualized approach that prioritizes inhaled glucocorticoids as the cornerstone of therapy, with escalation to combination inhaled glucocorticoid–long-acting β2-agonist regimens and consideration of additional agents for patients with more severe or uncontrolled disease (1, 2). Asthma management also incorporates nonpharmacologic interventions such as patient education and regular assessment of symptom control, risk factors, and inhaler technique. The primary goals of treatment are to achieve control of symptoms and reduce the risk of exacerbations, while keeping adverse drug effects minimal.

Major medication classes commonly used in the treatment of asthma and asthma exacerbations include

- Bronchodilators (beta-2 agonists, anticholinergics)
- Glucocorticoids
- Leukotrienreceptor antagonists
- Mast cell stabilizers
- Methylxanthines
- Biologic medications

Medications in these classes (see table Pharmacologic Treatment of Asthma) are inhaled, taken orally, or injected subcutaneously or intravenously; inhaled medications come in aerosolized and powdered forms. Use of aerosolized forms with a spacer or holding chamber facilitates deposition of the medication in the airways rather than the pharynx; patients are advised to wash and dry their spacers after each use to prevent bacterial contamination. In addition, use of aerosolized forms requires coordination between actuation of the inhaler (drug delivery) and inhalation; powdered forms reduce the need for coordination, but medication is delivered only when the patient fully inhales with a good effort.

TABLE

Pharmacologic Treatment of Asthma

Medication	Comments
Short-acting beta agonists	

Medication	Comments
Albuterol	Albuterol is used as a reliever* medication. It is not recommended for maintenance treatment. Regular use indicates diminishing asthma control and need for an additional medication. MDI-DPI is as effective as nebulized therapy if patients can coordinate the inhalation maneuver using the spacer and holding chamber. Nebulized albuterol can be mixed with other nebulizer solutions.
Levalbuterol	Levalbuterol is the *R*-isomer of albuterol. Levalbuterol may have fewer adverse effects.
Long-acting beta agonists (not to be used as monotherapy)	
Arformoterol	Arformoterol is the *R*-isomer of formoterol.
Formoterol	—
Salmeterol	Duration of action is 12 hours. One dose nightly is helpful for nocturnal asthma. Salmeterol is not to be used for acute symptom relief in an exacerbation.
Ultra–long-acting beta agonists (not to be used as monotherapy)	
Indacaterol	—
Olodaterol	—
Vilanterol	Vilanterol is available only in combination with fluticasone and/or umeclidinium.
Anticholinergics	

Medication	Comments
Ipratropium	Ipratropium may be mixed in the same nebulizer as albuterol. It should not be used as first-line therapy. Regular use provides no clear benefit for long-term maintenance therapy but should be added for treatment of acute symptoms.
Tiotropium	Tiotropium is longer acting than ipratropium. The lower dose SMI tiotropium is the only dose recommended for use in asthma.
Glucocorticoids (inhaled)	
Beclomethasone	Doses depend on severity and range from 1–2 puffs to whatever dose is needed to control asthma. May have systemic effects when used long term. Maximum threshold is that above which hypothalamic- pituitary-adrenal suppression is produced. If higher doses are necessary for asthma control, specialist consultation is recommended.
Budesonide	Doses depend on severity and range from 1–2 puffs to whatever dose is needed to control asthma. May have systemic effects when used long term. Maximum threshold is that above which hypothalamic- pituitary-adrenal suppression is produced. If higher doses are necessary for asthma control, specialist consultation is

Medication	Comments
	recommended.
Ciclesonide	Doses depend on severity and range from 1–2 puffs to whatever dose is needed to control asthma. May have systemic effects when used long term. Maximum threshold is that above which hypothalamic- pituitary-adrenal suppression is produced. If higher doses are necessary for asthma control, specialist consultation is recommended.
Flunisolide	Doses depend on severity and range from 1–2 puffs to whatever dose is needed to control asthma. May have systemic effects when used long term. Maximum threshold is that above which hypothalamic- pituitary-adrenal suppression is produced. If higher doses are necessary for asthma control, specialist consultation is recommended.
Fluticasone furoate	Doses depend on severity and range from 1–2 puffs to whatever dose is needed to control asthma. May have systemic effects when used long term. Maximum threshold is that above which hypothalamic- pituitary-adrenal suppression is produced. If higher doses are necessary for asthma control, specialist consultation is recommended.
Fluticasone propionate	Doses depend on severity and range from 1–2 puffs to whatever dose is

Medication	Comments
	needed to control asthma. May have systemic effects when used long term. Maximum threshold is that above which hypothalamic- pituitary-adrenal suppression is produced. If higher doses are necessary for asthma control, specialist consultation is recommended.
Mometasone	Doses depend on severity and range from 1–2 puffs to whatever dose is needed to control asthma. May have systemic effects when used long term. Maximum threshold is that above which hypothalamic- pituitary-adrenal suppression is produced. If higher doses are necessary for asthma control, specialist consultation is recommended.
Systemic glucocorticoids (oral)	
Methylprednisolone	Maintenance doses should be given in a single dose in the morning every day or every other day as needed for control. Short-course burst doses are effective for establishing control when initiating therapy or during a period of gradual deterioration. The burst should be continued until PEF = 80% of personal best or symptoms resolve, possibly requiring > 3–10 days of therapy.
Prednisolone	Maintenance doses should be given in a single dose in the morning every day or every other day as needed for control.

Medication	Comments
	Short-course burst doses are effective for establishing control when initiating therapy or during a period of gradual deterioration. The burst should be continued until PEF = 80% of personal best or symptoms resolve, possibly requiring > 3–10 days of therapy.
Prednisone	Maintenance doses should be given in a single dose in the morning every day or every other day as needed for control. Short-course burst doses are effective for establishing control when initiating therapy or during a period of gradual deterioration. The burst should be continued until PEF = 80% of personal best or symptoms resolve, possibly requiring > 3–10 days of therapy.
Combination medications	
Ipratropium/albuterol	Ipratroprium prolongs bronchodilator effect of albuterol.
Fluticasone/salmeterol	The medium-dose inhaler is indicated for asthma not controlled by low-to-medium doses of inhaled glucocorticoids. The high-dose is indicated for asthma not controlled by medium-to-high doses of inhaled glucocorticoids.
Budesonide/formoterol	The lower dose inhaler is indicated for asthma not controlled by low-to-medium doses of inhaled glucocorticoids. The higher dose inhaler is indicated for asthma not controlled by medium-to-high doses of inhaled glucocorticoids.
Mometasone/formoterol	The low-dose combination is recommended for asthma not controlled

Medication	Comments
	by low-dose glucocorticoids alone. The medium-dose combination is recommended for asthma not controlled by medium–dose inhaled glucocorticoids. The high-dose combination is recommended for asthma not controlled by high-dose inhaled glucocorticoids.
Fluticasone/vilanterol	Recommended starting dose is based on asthma severity.
Fluticasone/umeclidinium/vilanterol	Recommended starting dose is based on asthma severity.
Budesonide/ albuterol	Budesonide/albuterol is used as reliever therapy.
Mast cell stabilizers	
Cromolyn	Cromolyn (nebulized) should be taken before exercise or allergen exposure. One dose provides effective prophylaxis for 1–2 hours.
Nedocromil	Nedocromil is available as adjunct maintenance therapy for mild-to-moderate persistent asthma. It is less effective than inhaled glucocorticoids.
Leukotriene modifiers	
Montelukast	Montelukast is a leukotriene receptor antagonist that is a competitive inhibitor of leukotrienes D4 and E4. Caution must be exercised before initiating montelukast due to the potential risk of neuropsychiatric adverse drug events.
Zafirlukast	Zafirlukast is a leukotriene receptor antagonist that is a competitive inhibitor of leukotrienes D4 and E4.

Medication	Comments
	It must be taken 1 hour before or 2 hours after meals.
Zileuton	Zileuton inhibits 5-lipoxygenase. Dosing may limit adherence. Zileuton may cause liver enzyme elevations and inhibit metabolism of drugs processed by CYP3A4, including theophylline. Zileuton can be used in children, but only in those ≥ 12 years.
Methylxanthines	
Theophylline	The wide variability in metabolic clearance, drug interaction, drug-induced relaxation of the lower esophageal sphincter worsening reflux and worsening bronchospasm, and potential for adverse effects mandate routine serum level monitoring. Availability of safer alternatives has led to declining use of this medication.
Biologics	
Benralizumab	Benralizumab is used as an add-on treatment for patients with the eosinophilic phenotype.
Dupilumab	The initial (loading) dose should be given as 2 injections. Dupilumab is used as an add-on treatment for patients with the eosinophilic phenotype.
Mepolizumab	—
Omalizumab or biosimilar (omalizumab-igec)	—
Reslizumab	—
Tezepelumab	Tezepelumab is an add-on treatment for

Medication	Comments
	patients with severe, persistent asthma.

* Relievers, also called rescue inhalers, include a short-acting beta-2-agonist, an inhaled glucocorticoid plus a short-acting beta-2-agonist, or an inhaled glucocorticoid plus a long-acting beta-2-agonist.

DPI = dry-powder inhaler; MDI = metered-dose inhaler; SMI = soft mist inhaler; PEF = peak expiratory flow.

Adapted from the National Heart, Lung, and Blood Institute: Expert Panel Report 3, Guidelines for the diagnosis and management of asthma—full report 2007. August 28, 2007. Available at www.nhlbi.nih.gov/guidelines/asthma/asthgdln.pdf; and Expert Panel Working Group of the National Heart, Lung, and Blood Institute (NHLBI) administered and coordinated National Asthma Education and Prevention Program Coordinating Committee (NAEPPCC), Cloutier MM, Baptist AP, et al: 2020 Focused Updates to the Asthma Management Guidelines: A Report from the National Asthma Education and Prevention Program Coordinating Committee Expert Panel Working Group. *J Allergy Clin Immunol* 146(6):1217–1270, 2020. doi: 10.1016/j.jaci.2020.10.003

For additional information, see Global Initiative for Asthma. Global Strategy for Asthma Management and Prevention, 2025. Updated May 2025. Accessed May 20, 2025. Available from www.ginasthma.org

Beta agonists

Beta2-adrenergic receptor agonists and non-selective beta-agonists (eg, epinephrine, norepinephrine) relax bronchial smooth muscle, decrease mast cell degranulation and histamine release, inhibit microvascular leakage into the airways, and increase mucociliary clearance. Beta-2 agonist preparations may be short-acting, long-acting, or ultra–long-acting (see tables Pharmacologic Treatment of Asthma and Pharmacologic Treatment of Asthma Exacerbations).

Short-acting beta-2 agonists (eg, albuterol) 2 puffs every 4 hours inhaled may be used for relieving acute bronchoconstriction, and 2 puffs 15 to 30 minutes before exercise is used for preventing exercise-induced bronchoconstriction. There is a strong recommendation against the use of short-acting beta-2-agonist monotherapy for all patients with asthma (1), because monotherapy without the addition of an inhaled glucocorticoid has led to significantly more exacerbations, asthma-related emergency department visits, and worse lung function outcomes. Short-acting beta-2 agonists take effect within minutes and are active for up to 6 to 8 hours, depending on the medication. Tachycardia and tremor are the most common acute adverse effects of inhaled beta agonists and are dose-related. Mild hypokalemia occurs uncommonly. Use of levalbuterol (a solution containing the *R*-isomer of albuterol) theoretically minimizes adverse effects, but its long-term efficacy and safety are unproved. Oral beta agonists have more systemic effects and generally should be avoided.

Long-acting beta agonists (eg, salmeterol) are active for up to 12 hours. They are used in addition to inhaled glucocorticoids for moderate and severe asthma, and should never be used as monotherapy. Furthermore, they should not be used for acute symptom relief in an exacerbation. The dose is 1 to 2 puffs every 12 hours (except for formoterol, which in combination with an inhaled glucocorticoid may be dosed every 3 to 4 hours according to some protocols). For the prevention (and not the treatment) of exercise-induced bronchoconstriction, they may be administered along with glucocorticoids and taken 30 to 60 minutes before exercise. Long-acting beta agonists interact synergistically with inhaled glucocorticoids and permit lower dosing of glucocorticoids.

Ultra-long-acting beta agonists (eg, olodaterol, indacaterol) are active for up to 24 hours and, as with long-acting beta agonists, are used for moderate to severe asthma. Ultra-long-acting beta agonists should never be used as a monotherapy. They interact synergistically with inhaled glucocorticoids and permit lower dosing of glucocorticoids.

The safety of regular long-term use of beta agonists has been confirmed by multiple randomized trials and meta-analyses (3). Because the safety and efficacy of long-acting beta agonists have been demonstrated only when used in combination with an inhaled glucocorticoid, all long-acting and ultra-long beta agonists should be used only in combination with an inhaled glucocorticoid for patients whose condition is not adequately controlled with other asthma controllers (eg, low- to medium-dose inhaled glucocorticoids) or whose disease severity clearly warrants additional maintenance therapies. Daily use or diminishing effects of short-acting beta agonists or the use of ≥ 1 canister per month suggests inadequate control and the need to begin or intensify other therapies.

Anticholinergics (antimuscarinics)

Anticholinergics relax bronchial smooth muscle through competitive inhibition of muscarinic (M3) cholinergic receptors. Ipratropium and tiotropium are examples of anticholinergics. They may have an additive effect when combined with short-acting beta-2 agonists. The dose of ipratropium is 2 puffs every 6 hours as needed (maximum 12 puffs/day). Adverse effects include pupillary dilation, blurred vision, and dry mouth. Tiotropium soft mist inhaler is a 24-hour inhaled anticholinergic that can be used for patients with asthma. The dose is 2 puffs once a day (maximum 2 puffs/day). In patients with asthma, clinical trials of tiotropium added to either inhaled glucocorticoids or to a combination of an inhaled long-acting beta-2 agonist plus an inhaled glucocorticoid showed improved pulmonary function and decreased asthma exacerbations compared to placebo (4).

Inhaled combination therapy with glucocorticoids, long-acting beta-2 agonists, and long-acting muscarinic antagonists are recommended for the maintenance treatment of asthma for severe persistent asthma (step 5) in adults (1). Glucocorticoids provide anti-inflammatory effects, muscarinic antagonists provide bronchodilation via muscarinic receptor antagonism, and beta-2 agonists provide smooth muscle relaxation through beta-2 agonism. Fluticasone furoate/umeclidinium/vilanterol is used in the United States. Similar agents are used elsewhere; 2 combination inhalers, one containing mometasone furoate, indacaterol acetate, and glycopyrronium bromide and the other containing beclomethasone dipropionate, formoterol fumarate, and glycopyrronium bromide are available in the European Union. With the exception of the compound containing beclomethasone, which is administered twice daily, these agents are administered as one single inhalation daily.

Glucocorticoids

Glucocorticoids inhibit airway inflammation, reverse beta-receptor down-regulation, and inhibit cytokine production and adhesion protein activation. They block the late response (but not the early response) to inhaled allergens. Routes of administration include oral, IV, and inhaled. In acute asthma exacerbations, early use of systemic glucocorticoids often aborts the exacerbation, decreases the need for hospitalization, prevents relapse, and speeds recovery. Oral and IV routes are equally effective.

Inhaled glucocorticoids, also referred to as inhaled corticosteroids (ICS), are generally indicated for long-term suppression, control, and reversal of inflammation and symptoms. There is substantial evidence that the preferred reliever across all steps is a low-dose ICS–beta-agonist combination, used as needed for mild asthma and as both maintenance and reliever therapy for moderate-to-severe persistent asthma (5, 6). Specifically, clinical trials comparing inhaled glucocorticoid–beta-agonists to beta-agonist monotherapy demonstrated a substantial reduction in severe exacerbations compared to beta-agonist-only regimens, with similar or improved symptom control and lung function outcomes. Initiating as-needed inhaled glucocorticoid–beta-agonist therapy in patients with mild asthma effectively provides symptom relief and reduces the risk of severe asthma exacerbations, eliminating the necessity for daily maintenance treatment (7). Patients who adhere to inhaled glucocorticoid regimens also substantially reduce their need for maintenance oral glucocorticoid therapy.

Adverse local effects of inhaled glucocorticoids include dysphonia and oral candidiasis, which can be prevented or minimized by having the patient use a spacer, gargle with water after glucocorticoid inhalation, or both. Systemic effects are all dose related, can occur with oral or inhaled forms, and when due to inhaled forms, occur mainly with doses > 800 mcg/day. They include suppression of the adrenal-pituitary axis, osteoporosis, cataracts, skin atrophy, hyperphagia, and easy bruisability. Whether inhaled glucocorticoids suppress growth in children is unclear. Most children treated with inhaled glucocorticoids eventually reach their predicted adult height. Latent tuberculosis may be reactivated by systemic glucocorticoid use.

Mast cell stabilizers

Mast cell stabilizers inhibit histamine release from mast cells, reduce airway hyperresponsiveness, and block the early and late responses to allergens. They are given by inhalation prophylactically to patients with exercise-induced bronchoconstriction or allergen-induced asthma. They are ineffective once symptoms have occurred. They are the safest of all antiasthmatic medications but the least effective. They are not recommended for maintenance therapy of asthma in the United States.

Leukotriene receptor antagonists

Leukotriene receptor antagonists are taken orally and can be used for long-term control and prevention of symptoms in patients with mild persistent to severe persistent asthma. The main adverse effect is liver enzyme elevation (which occurs primarily with zileuton). Although rare, patients have developed a clinical syndrome resembling eosinophilic granulomatosis with polyangiitis.

The potential for severe neuropsychiatric reactions with montelukast has led to restrictions on its use and a boxed label warning in the United States in 2020 (8). These reports have led to concerns that the benefits of montelukast may not outweigh the risks in some patients, especially when symptoms are mild and can be adequately managed with other conventional therapies. Large-scale observational studies published after 2020 have provided mixed results on this potential association. For example, one nationwide cohort study from Sweden of children aged 6 to 17 years who used montelukast between 2007 and 2021 did not detect an association between neuropsychiatric adverse events and montelukast use (9). However, in a large systematic review of children and adults with asthma, montelukast use was not associated with depressive or suicidal symptoms, but older adults were found to be particularly susceptible to developing anxiety and sleep disorders (10). Additional longitudinal post-marketing surveillance studies are needed to make more definitive conclusions. Thus, in the context of asthma maintenance therapy, montelukast should only be prescribed after careful consideration of the potential risk of neuropsychiatric adverse drug events and a thorough discussion with the patient.

Methylxanthines

Methylxanthines (eg, theophylline) relax bronchial smooth muscle (probably by inhibiting phosphodiesterase) and may improve myocardial and diaphragmatic contractility; it is hypothesized that these effects may be related to their action on adenosine receptors and the inhibition of release of intracellular calcium. They also decrease microvascular leakage into the airway mucosa and inhibit the late response to allergens. They decrease the infiltration of eosinophils into the bronchial mucosa and T cells into respiratory epithelium.

Theophylline is used for long-term control as an adjunct to beta-2 agonists. Sustained-release oral theophylline helps manage nocturnal asthma. Theophylline has fallen into disfavor because of its many adverse effects and interactions compared with other medications. Adverse effects include headache, vomiting, cardiac arrhythmias, seizures, and aggravation of gastroesophageal reflux (by reducing lower esophageal sphincter pressure).

Methylxanthines have a narrow therapeutic index; multiple medications (any metabolized by the cytochrome P-450 pathway, eg, macrolide antibiotics) and conditions (eg, fever, liver disease, heart failure) alter methylxanthine metabolism and elimination. Serum theophylline levels should be monitored periodically and maintained between 5 and 15 mcg/mL (28 and 83 micromole/L).

Biologic medications

Biologic medications used in the treatment of asthma include agents that are anti-IgE antibodies (eg, omalizumab), antibodies to interleukin (IL)-5 (eg, benralizumab, mepolizumab, reslizumab), a monoclonal antibody that blocks IL-4 receptor-alpha to inhibit IL-4 and IL-13 signaling (dupilumab), and one that blocks thymic stromal lymphopoietin (TSLP) (tezepelumab). Biologics are used for the management of severe asthma refractory to a combination of step-up asthma therapies, usually high-dose inhaled glucocorticoids with a long-acting beta2-adrenergic receptor agonist, and primarily characterized by elevated allergic inflammation biomarkers (serum IgE, blood eosinophil count). Medication selection should be individualized for each patient's clinical scenario based on route, frequency, cost, and comorbid atopic disease. For example, in a patient with atopic dermatitis and asthma, dupilumab could be considered because it is also used in patients with atopic dermatitis.

Omalizumab or omalizumab-igec is indicated for patients with severe, allergic asthma who have elevated IgE levels. These medications may decrease asthma exacerbations, glucocorticoid requirements, and symptoms. Dosing is determined by a dosing chart based on the patient’s weight and IgE levels. The medication is administered subcutaneously every 2 to 4 weeks.

Mepolizumab, reslizumab, and benralizumab were developed for use in patients with eosinophilic (T2-high) asthma and are monoclonal antibodies that block IL-5 or its receptor, IL-5R. IL-5 is a cytokine that promotes eosinophilic inflammation in the airways. An ultra-long-acting anti-IL-5 monoclonal antibody depot preparation, depemokimab, intended to be administered subcutaneously every 6 months as maintenance therapy in severe eosinophilic asthma, has reduced asthma exacerbations and hospitalization rates in clinical trials (11).

Mepolizumab reduces exacerbation frequency, decreases asthma symptoms, and reduces the need for systemic glucocorticoid therapy in patients with asthma who depend on chronic systemic glucocorticoid therapy. Based on data from clinical trials, efficacy occurs with blood absolute eosinophil counts > 150/microL (0.15×10^9/L) (12). In patients requiring chronic systemic glucocorticoid therapy, the threshold for efficacy is unclear due to the suppressive effects of glucocorticoids on blood eosinophil counts yet mepolizumab has been shown to reduce or eliminate the need for systemic glucocorticoid therapy. Mepolizumab is administered subcutaneously every 4 weeks.

Reslizumab, a monoclonal antibody that binds to IL-5 receptors, also appears to reduce frequency of exacerbations and decrease asthma symptoms; however, it is administered intravenously every 4 weeks and therefore is used less often.

Benralizumab is a monoclonal antibody that binds to IL-5 receptors. It is indicated for the add-on maintenance treatment of severe asthma in patients aged 6 years or older with an eosinophilic phenotype. It has been shown to decrease exacerbation frequency and reduce and/or eliminate oral glucocorticoid use. It is administered as a subcutaneous loading dose followed by maintenance doses once every 8 weeks.

Dupilumab is a monoclonal antibody that blocks the IL-4R-alpha subunit, thereby simultaneously inhibiting IL-4 and IL-13 signaling. It is indicated for add-on maintenance treatment of patients with moderate-to-severe asthma aged 6 years or older with an eosinophilic phenotype or with oral glucocorticoid–dependent asthma. It is administered as a subcutaneous loading dose followed by maintenance dosing. The higher dosage is recommended for patients requiring concomitant oral glucocorticoids for whom dose reductions and cessation of systemic glucocorticoids is a goal.

Tezepelumab is a monoclonal antibody that binds specifically to TSLP and prevents its interaction with the TSLP receptor. It is prescribed as an add-on maintenance treatment for patients aged 12 years or older with severe asthma. Tezepelumab has been shown to significantly reduce asthma exacerbations compared with placebo (13). It is administered subcutaneously once every 4 weeks.

Clinicians who give any of these biologic medications should be prepared to identify and treat anaphylaxis or allergic hypersensitivity reactions. Anaphylaxis may occur after any dose of dupilumab, benralizumab, omalizumab, tezepelumab, or reslizumab even if previous doses have been well tolerated. Allergic hypersensitivity reactions have been reported with mepolizumab. Mepolizumab use has been associated with herpes zoster infection; therefore, zoster vaccination is recommend prior to initiation of therapy unless contraindicated. Pharyngitis and arthralgias have been reported with tezepelumab.

Pearls & Pitfalls

- Prepare for possible anaphylactic or hypersensitivity reactions in patients being treated with omalizumabor omalizumab-igec, mepolizumab, reslizumab, benralizumab, tezepelumab, or dupilumab regardless of how such treatments have been tolerated previously.

Other therapeutic approaches

Other medications are used in asthma treatment uncommonly and in specific circumstances. Magnesium is often used in the emergency department for the treatment of acute exacerbations, but it is not recommended in the management of chronic asthma.

Allergen immunotherapy may be indicated when symptoms are triggered by allergy, as suggested by history and confirmed by allergy testing. Immunotherapy refers to the controlled, repeated administration of specific allergens to induce immunologic tolerance, primarily through the modulation of IgE-mediated immune responses. It consists of a build-up (where doses are systematically increased as tolerated based on established protocols) and maintenance phase (where the highest tolerated dose is administered). If symptoms are not significantly relieved after 3 to 5 years of maintenance therapy, then therapy is stopped. The optimum duration of maintenance therapy is unknown.

References

Drug Information for the Topic

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Treatment of Acute Asthma Exacerbations

*Full Review: Dec 2025 By **Victor E. Ortega**, MD, PhD, Mayo Clinic | **Sergio E. Chiarella**, MD, Mayo Clinic | Peer reviewed by **M. Patricia Rivera**, MD, University of Rochester Medical Center*
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Asthma is a common chronic respiratory disease that is noncommunicable. According to the Global Initiative for Asthma, asthma is a heterogenous disease and usually characterized by chronic airway inflammation (1). Uncontrolled asthma is associated with a higher risk of exacerbations requiring escalation of therapy. The goal of asthma exacerbation treatment is to relieve symptoms and return patients to their optimal lung function. Treatment includes:

- Inhaled bronchodilators (beta-2 agonists and anticholinergics)
- Usually systemic glucocorticoids
- Sometimes magnesium

(See also Asthma and Pharmacologic Treatment of Asthma.)

Patients having an asthma exacerbation are instructed to self-administer 2 to 4 puffs of inhaled albuterol or a similar short-acting beta agonist up to 3 times spaced 20 minutes apart for an acute exacerbation and to measure peak expiratory flow (PEF) if possible. When these short-acting rescue medications are effective (symptoms are relieved and PEF returns to > 80% of baseline), the acute exacerbation may be managed in the outpatient setting. Patients who do not respond, have severe symptoms, or have a PEF persistently < 80% should follow a treatment management program outlined by the physician or should go to the emergency department (for specific dosing information, see table Pharmacologic Treatment of Asthma Exacerbations).

TABLE

Pharmacologic Treatment of Asthma Exacerbations*, †

Medication	Form	Dosage in Children	Dosage in Adults	Comments
Short-acting beta-2 agonists‡				
Albuterol	HFA: 90 mcg/puff	Same as adults	4–10 puffs every 20 minutes for 3 doses, then up to 6–10 doses	MDI is as effective as nebulized solution if patients can coordinate

Medication	Form	Dosage in Children	Dosage in Adults	Comments
			every 1–2 hours	inhalation maneuver using spacer and holding chamber.
	Nebulized solution: 5 mg/mL and 0.63, 1.25, and 2.5 mg/3 mL	0.15 mg/kg (minimum 2.5 mg) every 20 minutes for 3 doses, then 0.15–0.3 mg/kg up to 10 mg every 1–4 hours as needed Alternatively, 0.5 mg/kg/hour continuous nebulization (up to 10–20 mg per hour in children < 12 years) In children > 12 years, up to 10–15 mg/hour	2.5–5 mg every 20 minutes for 3 doses, then 2.5–10 mg every 1–4 hours as needed Alternatively, 10–15 mg/hour continuous nebulization	Continuous nebulization is similarly effective compared to episodic nebulization, but it increases frequency of adverse effects.
Levalbuterol	HFA: 45 mg/actuation	Same as adults	4–8 puffs every 20 minutes for 3 doses, then every 1–4 hours as needed	Levalbuterol is the *R*-isomer of albuterol. 0.63 mg is equivalent to 1.25 mg racemic albuterol. Levalbuterol may have fewer adverse

Medication	Form	Dosage in Children	Dosage in Adults	Comments
				effects than albuterol.
	Nebulized solution: 0.31, 0.63, 1.25 mg/3 mL, and 1.25 mg/0.5 mL	0.075 mg/kg (minimum 1.25 mg) every 20 minutes for 3 doses, then 0.075–0.15 mg/kg up to 5 mg every 1–4 hours as needed	1.25–2 mg every 20 minutes for 3 doses, then 1.25–5 mg every 1–4 hours as needed Alternatively, 5–7.5 mg/hours continuous nebulization	—
Anticholinergics				
Ipratropium	Nebulized solution: 500 mcg/2.5 mL (0.02%) MDI: 18 mcg per puff	0.25–0.5 mg every 20 minutes for 3 doses, then every 2–4 hours as needed Children: 4–8 puffs every 20 minutes as needed for up to 3 hours Adolescents: 8 puffs every 20 minutes for up to 3 hours	0.5 mg every 20 minutes for 3 doses, then every 2–4 hours as needed	Ipratropium should be added to beta-2 agonists and not used as first-line therapy. It may be mixed in same nebulizer as albuterol. Dose delivered from MDI is low and has not been studied in exacerbations.
Combination medications				
Ipratropium/albuterol	SMI: 20 mcg ipratropium	—	1 puff every 30 minutes for 3 doses, then	Ipratropium prolongs bronchodilator

Medication	Form	Dosage in Children	Dosage in Adults	Comments
	and 100 mcg albuterol/puff		every 2–4 hours as needed	effect of albuterol.
	Nebulized solution: 0.5 mg ipratropium and 2.5 mg albuterol in a 3-mL vial	1.5 mL every 20 minutes for 3 doses, then up to 3 hours as needed	3 mL every 30 minutes for 3 doses, then every 2–4 hours as needed	—
Budesonide/formoterol	HFA: 80 or 160 mcg budesonide and 4.5 mcg formoterol	4 to < 12 years Budesonide 80 mcg/formoterol 4.5 mcg 1 puff as needed for symptom control administered after a few minutes as needed Maximum daily dose including maintenance and reliever therapy is 8 puffs ≥ 12 years: Budesonide 160 mcg formoterol 4.5 mcg 1 to 2 puffs as needed for symptom relief after a few minutes as needed to	2 puffs twice a day as needed Maximum total daily maintenance and rescue (reliever) of 12 puffs (54 mcg)	Additional daily maintenance therapy should be continued even if used as reliever therapy.

Medication	Form	Dosage in Children	Dosage in Adults	Comments
		control symptoms Maximum daily dose is 12 puffs		
Budesonide/albuterol	HFA: 80 mcg budesonide/90 mcg albuterol	< 18 years: not used	≥ 18 years: 80 mcg budesonide/90 mcg albuterol 2 puffs as needed Maximum 6 doses or 12 puffs in 24-hour period	—
Systemic glucocorticoids				
Methylprednisolone	Tablets: 2, 4, 8, 16, and 32 mg	0–11 years of age: 1–2 mg/kg once a day or in 2 divided doses until symptoms resolve and PEF is at least 80 percent of patient's personal best	40–80 mg day or in 2 divided doses until symptoms resolve and PEF is at least 70 % of patient's personal best	IV has no advantage over oral administration if gastrointestinal function is normal. Higher doses provide no advantage in severe exacerbations. The dose is administered once or twice a day until FEV1 or PEF = 50% of predicted or personal best and then lower

Medication	Form	Dosage in Children	Dosage in Adults	Comments
				the dose to twice a day, usually within 48 hours. Therapy after a hospitalization or ED visit may last 5–10 days in adults and 3–5 days in children. Tapering the dose is not needed if patients are also given inhaled glucocorticoids.
	Injectable suspension: 20 mg/mL, 40 mg/mL, and 80 mg/mL Solution for injection: 40 mg, 125 mg, 500 mg, 1000 mg, and 2000 mg	0–11 years: 1–2 mg/kg once a day or in 2 divided doses until symptoms resolve and PEF is at least 70% of patient's personal best	60–80 mg in 3 or 4 divided doses for 48 hours, then 30–40 mg/day until PEF reaches 70% of personal best	—
Prednisolone	Tablets: 5 mg Orally disintegrating tablets: 10, 15, and 30 mg	1–2 mg/kg in 1–2 divided doses until PEF is 70% of predicted or personal best	40 to 80 mg once a day or in 2 divided doses until PEF reaches 70% of patient's predicted or personal best	Maximum dose: < 2 years: 20 mg/day 3–5 years: 30 mg/day 6–11 years: 40 mg/day

Medication	Form	Dosage in Children	Dosage in Adults	Comments
	Solution: 5, 10, 15, 20 and 25 mg/5 mL	Same as above	Same as above	Same as above
Prednisone	Tablets: 1, 2.5, 5, 10, 20, and 50 mg	1–2 mg/kg daily for 5 days	40–60 mg/day in single daily dose or divided every 12 hours for 3–10 days	Maximum dose: < 2 years: 20 mg/day 3–5 years: 30 mg/day 6–11 years: 40 mg/day
	Concentrate: 5 mg/mL Solution: 5 mg/5 mL	Same as above	Same as above	Same as above

* All ages unless specified differently.

† Amount and timing of ongoing doses are dictated by clinical response.

‡ Short-acting-beta2-agonists must be used in combination with an inhaled corticosteroid (glucocorticoid).

ED = emergency department; FEV1= forced expiratory volume in 1 second; HFA = hydrofluoroalkane; MDI = metered-dose inhaler; PEF = peak expiratory flow; SMI = soft mist inhaler.

Adapted from National Heart, Lung, and Blood Institute: Expert Panel Report 3: Guidelines for the diagnosis and management of asthma—full report 2007. August 28, 2007. Available at www.nhlbi.nih.gov/guidelines/asthma/asthgdln.pdf; and Expert Panel Working Group of the National Heart, Lung, and Blood Institute (NHLBI) administered and coordinated National Asthma Education and Prevention Program Coordinating Committee (NAEPPCC), Cloutier MM, Baptist AP, et al: 2020 Focused Updates to the Asthma Management Guidelines: A Report from the National Asthma Education and Prevention Program Coordinating Committee Expert Panel Working Group. *J Allergy Clin Immunol* 146(6):1217–1270, 2020. doi: 10.1016/j.jaci.2020.10.003.

For additional information, see Global Initiative for Asthma. Global Strategy for Asthma Management and Prevention, 2025. Updated May 2025. Accessed May 20, 2025. Available from www.ginasthma.org.

Emergency department care

Inhaled bronchodilators (beta-2 agonists and anticholinergics) are the mainstay of asthma treatment in the emergency department. In adults and older children, albuterol given by a metered-dose inhaler (MDI) and spacer is as effective as that given by nebulizer. Nebulized treatment is preferred for younger children because of difficulties coordinating MDIs and spacers. It should be emphasized that, contrary to popular belief, no data favor continuous beta-2 agonist nebulization over intermittent administration; continuous nebulization is inconsistent with the pharmacokinetics of beta-agonists binding to their receptors. Evidence suggests that bronchodilator response improves when the nebulizer is powered with a mixture of helium and oxygen (heliox) rather than with oxygen. Given its lower density, helium is thought to assist with delivery of bronchodilators to distal airways. However, technical aspects of using helium for nebulization (availability, calibration of helium concentration, need for custom masks to avoid dilution with room air) have limited its widespread acceptance.

Nebulized ipratropium can be co-administered with nebulized albuterol for patients who do not respond optimally to albuterol alone; some evidence favors simultaneous high-dose beta-2 agonist and ipratropium as first-line treatment.

Systemic glucocorticoids (prednisone, prednisolone, methylprednisolone) should be given for all but the mildest acute exacerbation; they are unnecessary for patients whose PEF normalizes after 1 or 2 bronchodilator doses. IV and oral routes of administration are probably equally effective. IV methylprednisolone can be given if an IV line is already in place and can be switched to oral dosing whenever necessary or convenient. In general, higher doses (prednisone 50 to 60 mg once a day) are recommended for the management of more severe exacerbations requiring in-patient care while lower doses (40 mg once a day) are reserved for outpatient treatment of milder exacerbations. Although evidence about optimal dose and duration is weak, a treatment duration of 3 to 5 days in children and 5 to 7 days in adults is recommended as adequate by most guidelines and should be tailored to the severity and duration of an exacerbation (1, 2).

Intravenous magnesium sulfate is an adjunct in the treatment of acute asthma exacerbations that are severe or life-threatening and unresponsive to initial therapy with inhaled beta-2 agonists and systemic glucocorticoids. Evidence from meta-analyses demonstrates that IV magnesium sulfate may lead to a modest decrease in hospital admissions and improvement in lung function, particularly in adults and children with more severe presentations (3, 4). Magnesium sulfate relaxes smooth muscle, but efficacy in management of asthma exacerbation in the emergency department is debated.

Antibiotics are indicated only when history, examination, or chest radiograph suggests underlying bacterial infection; most infections underlying asthma exacerbations are probably viral in origin.

Supplemental oxygen is indicated for hypoxemia and should be given by nasal cannula or face mask at a flow rate or concentration sufficient to maintain oxygen saturation > 90%.

Subcutaneous epinephrine is not recommended as a routine therapy for asthma exacerbations and should be reserved for specific indications such as concomitant anaphylaxis or refractory, life-threatening asthma when other therapies, such as inhaled short-acting beta-2-agonists, are not available or effective. In such cases, intramuscular or subcutaneous epinephrine may be administered at a dose of 0.3 to 0.5 mg (1 mg/mL solution for adults), not to exceed 0.5 mg per injection and repeated every 5 to 10 minutes as needed. In prepubertal children the dose is 0.01 mg/kg every 5-20 minutes for 3 doses (maximum dose 0.3 mg) administered intramuscularly (5).

Terbutaline is an alternative to epinephrine for children. Terbutaline may be preferable to epinephrine because of fewer cardiovascular effects and longer duration of action, but supplies are limited.

Theophylline has very little role in treatment of an acute asthma exacerbation. It is associated with an increased risk of adverse effects such as arrhythmias and seizures. As a result, theophylline is not routinely recommended for acute asthma exacerbations and is reserved for selected cases, such as for patients who do not respond to standard therapies and are hospitalized with severe or life-threatening exacerbations (1).

Reassurance is the best approach when anxiety is the cause of asthma exacerbation. Anxiolytics and morphine are relatively contraindicated because they are associated with respiratory depression, and morphine may cause anaphylactoid reactions due to release of histamine by mast cells; these medications may increase mortality and the need for mechanical ventilation.

Hospitalization

Hospitalization generally is required if patients have not returned to their baseline within 4 hours of aggressive emergency department treatment. Criteria for hospitalization vary, but definite indications are:

- Failure to improve
- Worsening fatigue
- Relapse after repeated beta-2 agonist therapy
- Significant decrease in PaO_2 (to < 50 mm Hg)
- Significant increase in $PaCO_2$ (to > 40 mm Hg)

A significant increase in $PaCO_2$ indicates progression to respiratory failure.

Noninvasive positive pressure ventilation (NIPPV) may be needed in patients whose condition continues to deteriorate despite aggressive treatment to alleviate the work of breathing. NIPPV includes expiratory positive airway pressure that reduces the effect of intrinsic positive pressure that remains in the alveoli at the end of expiration during mechanical ventilation. It also includes inspiratory positive airway pressure that reduces the effects of increased airway resistance. Providing pressure support during both inspiratory and expiratory phases helps to reduce the work of breathing.

Endotracheal intubation and invasive mechanical ventilation may be needed for respiratory failure. NIPPV can be used to prevent intubation if used early in the course of a severe exacerbation and should be considered in patients with acute respiratory distress with a level of $PaCO_2$ that is inappropriately high in relation to the degree of tachypnea. It should be reserved for exacerbations that, despite immediate therapy with bronchodilators and systemic glucocorticoids, result in respiratory distress, using criteria such as tachypnea (respiratory rate > 25 per minute), use of accessory respiratory muscles, $PaCO_2$ > 40 but < 60 mm Hg, and hypoxemia. Mechanical ventilation should be used rather than NIPPV if patients have any of the following:

- $PaCO_2$ > 60 mm Hg
- Decreased level of consciousness
- Excessive respiratory secretions
- Facial abnormalities (ie, surgical, traumatic) that could impede noninvasive ventilation

Mechanical ventilation should be strongly considered if there is no convincing improvement after 1 hour of NIPPV.

Intubation and mechanical ventilation allow the provision of sedation to further alleviate the work of breathing, but the routine use of neuromuscular blocking agents should be avoided because of possible interactions with glucocorticoids that can cause prolonged neuromuscular weakness. Ketamine may be used for awake intubation if the user has familiarity with its use and adverse effects (eg, laryngospasm, rigidity, bronchorrhea).

Generally, volume-cycled ventilation in assist-control mode is used because it provides constant alveolar ventilation when airway resistance is high and changing. The ventilator should be set to a relatively low frequency with a relatively high inspiratory flow rate (> 80 L/minute) to prolong exhalation time, minimizing auto-positive end-expiratory pressure (auto-PEEP). Initial tidal volumes can be set to 6 to 8 mL/kg of ideal body weight, and extrinsic PEEP should be used to facilitate patient-initiated triggering and minimize ventilator dyssynchrony from auto-PEEP. High peak airway pressures will generally be present because they result from high airway resistance and inspiratory flow rates. In these patients, peak airway pressure does not reflect the degree of lung distention caused by alveolar pressure. However, if plateau pressures exceed 30 to 35 cm water, then tidal volume should be reduced to limit the risk of pneumothorax. When reduced tidal volumes are necessary, a moderate degree of hypercapnia is acceptable ("permissive hypercapnia"), but if arterial pH falls below 7.10, a slow sodium bicarbonate infusion can be considered to maintain pH between 7.20 and 7.25, especially if there is hemodynamic instability. Once airflow obstruction is relieved and PaCO2 and arterial pH normalize, patients can usually be quickly liberated from the ventilator. (For further details, see Respiratory Failure and Mechanical Ventilation.)

Other therapy

Other therapies are reportedly effective for asthma exacerbation, but none have been thoroughly studied. A mixture of helium and oxygen (heliox) is used to decrease the work of breathing and improve ventilation through a decrease in turbulent flow attributable to helium, a gas less dense than oxygen. Despite the theoretical benefits of heliox, studies have reported conflicting results concerning its efficacy; lack of ready availability and inability to concurrently provide high concentrations of oxygen (due to the fact that 70 to 80% of the inhaled gas is helium) may also limit its use (6, 7).

Vocal fold dysfunction, also called inducible laryngeal obstruction, is characterized by the inappropriate adduction of the vocal folds during inspiration, which can lead to airflow obstruction and mimic an asthma exacerbation. Management of vocal fold dysfunction involves a multidisciplinary approach, including speech therapy to teach breathing techniques. Heliox could also be beneficial for the management of patients with vocal fold dysfunction.

General anesthesia with agents such as sevoflurane and isoflurane in patients with status asthmaticus causes bronchodilation by an unclear mechanism, perhaps by a direct relaxant effect on airway smooth muscle or attenuation of cholinergic tone.

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